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Secure speech retrieval method using deep hashing and CKKS fully homomorphic encryption

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Abstract
The development of deep learning technology makes speech retrieval and recognition more accurate and efficient. Meanwhile, the privacy leakage problem of speech data is becoming increasingly prominent, but the emergence of fully homomorphic encryption (FHE) technology can alleviate the concerns about privacy information. In order to protect the privacy of speech data and deep binary hash codes, and realize the privacy-preserving similarity calculation, a secure speech retrieval method using deep hashing and CKKS (Cheon-Kim-Kim-Song) FHE was proposed. Firstly, a speech CKKS FHE scheme is designed to encrypt the original speech data. Then, the spectrogram image features of the original speech data are extracted as the input of triplet convolutional neural network (Tri-CNN) to generate efficient and compact deep binary hash codes, which are encrypted and uploaded to the cloud together with the encrypted speech data. When retrieving, the deep binary hash codes of the querying speech is extracted, encrypted and sent to the cloud server as a search trapdoor, and the security similarity is calculated with the index sequence in the secure index table. The experimental results show that the mean average precision of the proposed method in the TIMIT and THCHS-30 data sets is more than 93%, with a loss of about 2% compared with the plaintext domain, but with higher security.

Keywords Speech retrieval · Triplet CNN · Secure index · CKKS fully homomorphic encryption · Spectrogram image features

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1 Introduction

With the continuous progress of cloud storage technology, more and more companies and individuals choose to store massive audio, image and other multimedia data in the cloud in order to save the storage space of local computers [1]. However, while cloud storage brings great convenience to people, it also causes frequent data leakage and other problems. Therefore, how to ensure the secure storage and secure and efficient retrieval of cloud data has great practical significance [2–4]. Speech has special semantic functions and plays an important role in applications such as meeting records, court evidence, and communication records. Usually, in order to meet the needs of user privacy protection, private speech data will be encrypted before outsourcing, but it will inevitably bring certain difficulties to the rapid retrieval of data. Therefore, how to realize secure and efficient multimedia data retrieval is also an urgent problem to be solved [5, 6].

At present, the existing encrypted speech retrieval methods using perceptual hash [7] and biological hash [8] all use the existing manual perception features to construct the perceptual hash abstract, which has problems such as human subjectivity, large amount of computation, and failure to reflect the rich speech information and content perception information contained in the speech. With the wide application of deep learning technology in the field of speech retrieval and recognition, it provides a good idea to solve these problems. For example, speech retrieval methods such as deep hashing [9] and searchable encryption [10] all combine deep learning with deep hashing algorithm to extract deeper semantic features and reduce the dimension of the extracted depth features, significantly improving the efficiency and accuracy of speech retrieval. However, in order to ensure the efficiency and accuracy of speech retrieval, the existing encrypted speech retrieval methods usually only protect the privacy of the original speech data, and search and match the deep hashing index in the plaintext domain, which is easy to cause privacy information leakage. The existing secure speech retrieval methods using searchable encryption only support theoretical retrieval matching of security indexes (encrypted deep binary hash codes), and the derivation process is very complicated.

With the rise of multimedia secure retrieval technology with privacy protection [2–4, 6], FHE technology is used to calculate and process multimedia data features in the encrypted domain, which alleviates the concerns about privacy security. In addition, the existing speech retrieval methods using deep learning usually use a single label as the supervision information. During the model training process, the relative similarity between data cannot be obtained, which makes the generated hash codes suboptimal. In order to ensure the accuracy of speech retrieval, the deeper semantic information can only be contained through longer deep hashing codes, which seriously affects the retrieval efficiency and causes memory consumption. The mature triplet (anchor image, positive image, negative image) deep hashing method [11, 12] in the field of image retrieval and recognition can obtain relatively similar concepts between data, and can generate hash codes with both semantic information and category information, which provides technical support for efficient speech retrieval. The spectrogram [13] carries time domain and frequency domain information in a two-dimensional mode, which is a good representation of speech features. However, there are huge differences in the characteristics of the image field and the audio field. The original audio is a one-dimensional time series signal, which is essentially different from the two-dimensional image [14]. However, converting speech into the form of spectrogram images, and studying the encrypted speech retrieval method using deep learning from the perspective of spectrogram images, can bring the advantages of deep hashing

learning method in the image field into full play in the speech retrieval field, and has great research significance and application value for realizing the secure and efficient retrieval of massive speech data in the cloud.

In order to ensure the privacy security of cloud speech data and hash codes, and realize efficient secure speech retrieval under the premise of ensuring retrieval accuracy, a secure speech retrieval method using deep hashing and CKKS FHE is proposed. The main contributions of this paper are as follows:

1) Using Tri-CNN, a triplet deep hashing construction method suitable for spectrogram image features is designed, which can generate efficient and compact deep binary hash codes and improve the efficiency and accuracy of speech data securely retrieval.

2) A speech CKKS FHE scheme is designed using CKKS algorithm [15]. The computational complexity is reduced by using segmentation and batch processing, and the expansion of encrypted speech data is reduced by combining the modular switching and relinearization technology.

3) A secure speech retrieval method supporting the calculation of security similarity is proposed. The cloud server calculates the distance between secure indexes through the cosine similarity so as to search and match, which not only ensures the privacy protection of speech data, but also prevents the deep binary hash codes from being maliciously inferred, thus realizing the security sharing of speech data.

The rest of the paper is organized as follows: Section 2 introduces the related work. Section 3 presents the details of the secure speech retrieval method. Section 4 gives the experimental results and analysis, and compares the performance with different existing method. Section 5 concludes the work of this paper.

2 Related works

Since the rise of deep learning technology, a secure speech retrieval method using deep hashing has been proposed, which significantly improves the retrieval efficiency and accuracy. For example, Chen et al. [16] proposed an accurate and Scalable Identity-Oriented speech retrieval, which retrieved all speech clips uttered by the same speaker by calculating the cosine distance between the plaintext binary hash codes. Zhang et al. [17] proposed an encrypted speech retrieval algorithm based on CNN and deep hashing, which combines CNN and hash function to generate deep binary hash codes, and uses normalized Hamming distance for retrieval and matching. Zhang et al. [18] proposed speech keyword retrieval method using attention mechanism and multimodal information fusion to achieve cross modal retrieval of speech text by obtaining shared features of speech and text patterns. Yuan et al. [19] proposed a deep hashing network method based on attention, which can significantly improve the accuracy and speed of speech search in low resource scenarios using the query by example method. Through the analysis of the above reference, the existing secure speech retrieval technology based on deep hashing is searching and matching in the plaintext domain, which does not meet the requirements of user privacy protection and security sharing. In order to solve the serious problem of data security and the risk of privacy leakage, researchers proposed searchable encryption technology. Zhang et al. [20] proposed a secure speech retrieval scheme based on multiuser searchable encryption. By combining ciphertext-policy attribute-based encryption with searchable encryption to construct an encrypted speech library, and then using the random invertible matrix generated by pseudorandom functions to encrypt speech keywords to construct a

secure index. During retrieval, security similarity is calculated through Euclidean distance, which realizes data privacy protection and security sharing. Li et al. [21] proposed a voice searchable scheme based on granule computing technique, which uses obfuscation matrix and Advanced Encryption Standard (AES) algorithm to process voice features and original voice respectively and store them in the cloud, when retrieving the distance between the test granule vector and all granule vectors in the voice database can be calculated by k-nearest neighbor ciphertext granule search algorithm to achieve category retrieval. Yu et al. [10] proposed a searchable encryption scheme based on lattice proxy time control, which uses keyword based public key encryption to ensure the privacy and searchability of data, and introduced public key encryption with keywords search to achieve time controlled multi-user search scheme using proxy re-encryption technology. Li et al. [22] proposed a multi-user voice searchable encryption scheme for home IoT system, in which users used AES algorithm to encrypt voice data and extracted the feature of Mel Frequency Cepstrum Coefficient (MFCC) of the original voice as the keyword, then by adding noise and confusion matrix to encrypt features to achieve matching retrieval in encrypted domain, the security of user voice data is protected to the greatest extent. Although searchable encryption technology can better realize security sharing, at present, most methods are only theoretical derivation, and the construction process of secure index is too complicated.

In order to better realize the privacy protection and secure sharing of speech data in the cloud environment, researchers proposed the FHE technology that supports the computation in the encrypted domain. Rahulamathavan et al. [3] proposed an algorithm that uses FHE technology to verification the back-end of the speaker system, and used CKKS FHE scheme to obtain a real-time and non-interactive solutions. Zhang et al. [9] proposed a deep polynomial network, which can be used as an acoustic model to encrypt speech, and allow the client to use homomorphic encryption (HE) algorithm to ensure the confidential of data, at mean while DPN can still make frame-level predictions over the encrypted speech and return them in encrypted form. Chen et al. [23] studied a privacy protection technology in cloud environment, that is, the application of encrypted fragile audio watermarking technology in audio authentication system, and this scheme implement the computation of Mel-frequency cepstral coefficient in the encrypted domain with additive HE. Thaine et al. [24] described a method for extracting the Mel-frequency and Bark-frequency cepstral coefficient from the encrypted signal without having to decrypt any intermediate value, realizing secure end-to-end automatic speech recognition. Tang et al. [25] proposed a new scheme to realize the decompression of FLAC in the HE domain through the dynamic control and static control in the regularized FLAC decoding. Meftah et al. [26] designed an efficient deep convolution neural network based on FHE, and it is a batch neuron with low depth, which can simultaneously evaluate multiple quantized ReLU-activated neurons on encrypted data without approximation. Natarajan et al. [27] investigated a privacy-preserving variant of ideation detection application flow involving homomorphic evaluation of neural networks, which protected the confidential of private data. In order to achieve similarity measurement in the encrypted domain, Liu et al. [28] used CKKS FHE scheme to encrypt user data samples, and then used Euclidean distance, Pearson similarity, cosine similarity to implements a secure KNN classification scheme in cloud servers for Cyberspace.

Most of the existing encrypted speech retrieval methods do not consider the hash codes encryption, there is still a risk of privacy disclosure in the retrieval process, or a few of the secure retrieval schemes based on partial theoretical derivation of searchable encryption. To solve these problems, this paper presents a secure speech retrieval method based on deep hashing and CKKS FHE.

3 The proposed method

3.1 System model

Figure 1 shows the system model of secure speech retrieval method. The model consists of three entities: data owner (DO), data user (DU) and cloud server (CS).

As shown in Fig. 1, the division of labor of each entity is as follows:

1) Data owner (DO): Generates the public key (*pk*), private key (*sk*), and ciphertext calculation key (*evk*) locally, encrypts the local original speech data sets and its corresponding deep binary hash codes, obtains the encrypted speech data and its corresponding secure index, and store them in CS in one-to-one correspondence.
2) Cloud server (CS): CS provides remote storage and retrieval matching services for users. CS stores encrypted speech data and secure index table I uploaded by DO, calculating the similarity between each secure index in secure index table I and search trapdoor, and returns the search result to DU.
3) Data user (DU): After preprocessing the querying speech data, the DU uses the triplet deep hashing method based on the Tri-CNN to generate a deep binary hash code of the spectrogram image features, encrypts it and sends it to CS as search trapdoor for security similarity calculation, and finally decrypts the encrypted speech data returned by CS.

Since CS is "honest and inquisitive", transmission networks can also be maliciously attacked. Therefore, security problems that may occur when DO uploads encrypted speech data and secure index table I to the cloud server for storage, and when DU sends search trapdoor to the CS for secure speech retrieval are listed:

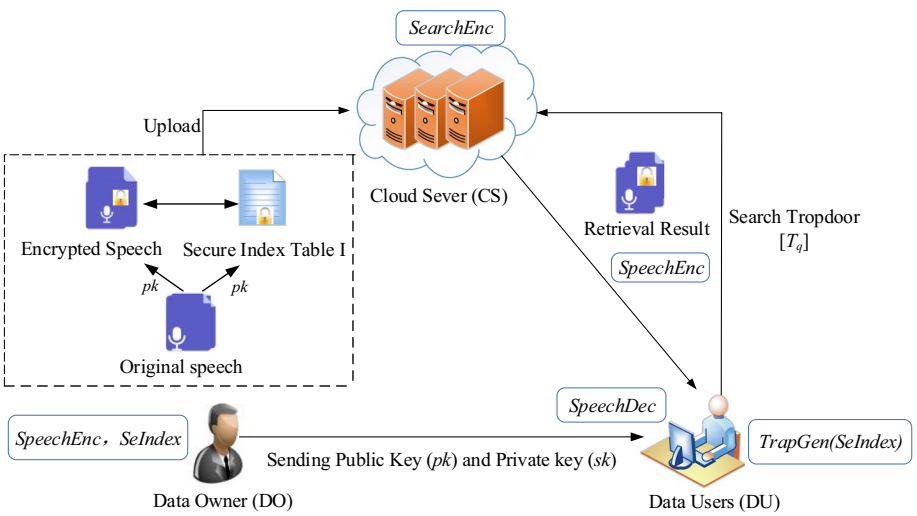


Fig. 1 Secure speech retrieval system model

1) CS strictly follows the designed protocol, but it can also infer other additional information of the original speech data through the information legally received in the retrieval process;
2) CS attempts to steal the user's *pk* and *sk*, and attempts to decipher the user's encrypted speech data and *sk* by relying on the stored encrypted speech data;
3) During transmission, the encrypted speech data uploaded by DO and the encrypted result returned by CS may be intercepted by hackers to decipher the user's speech sensitive data.

3.2 Description of secure speech retrieval algorithm

The secure speech retrieval method proposed in this paper mainly includes 8 algorithms: *Setup*, *Initialization Parameter*, *GenKey*, *SpeechEnc*, *SeIndex*, *TrapGen*, *SearchEnc*, and *SpeechDec* algorithms. Each algorithm is described as follows:

1) Pretreatment. $Swp \leftarrow Setup()$. Input speech data set and polynomial module to generate speech array $swp[b][z]$, z is the frame length, b is the data amount of each frame.
2) Initialization. $N,P,q,\Delta \leftarrow Initialization\ Parameter()$. The algorithm selects a power N of 2, ciphertext modulus q, scaling factor Δ, a base $p>0$ and modulus P (for rescaling), defines $Q=q_0 \cdot p^{\mathrm{L}}$ (q_0 is a complex modulus), and makes N,P,q,Δ,L meet the safety level coefficient λ, L represents the upper limit of depth. Then select a distribution X_s related to the private key, an error distribution X_e, and a random distribution X_r for encryption.
3) Key generation algorithm. $sk,pk,evk \leftarrow KenGen(\lambda)$. Instantiate $s \leftarrow X_s$, $a \leftarrow R_Q$, and $e \leftarrow X_e aa$, and set the private key to $sk \leftarrow (1,s)$, and calculate the public key $pk \leftarrow (b,a) \in R_Q^2$, where $b=-a \cdot s+e(\mathrm{mod}Q)$; Instantiate $a\prime \leftarrow R_{PQ}$, and $e \leftarrow X_e$, Set auxiliary calculation key $evk \leftarrow (b\prime,a\prime) \in R_{PQ}^2$, where $b\prime=-a\prime \cdot s+e\prime+P \cdot s^2(\mathrm{mod}P \cdot Q)$.
4) Speech encryption algorithm. $[C] \leftarrow SpeechEnc(pk,S)$. Enter public key *pk* and speech data $S=\{S_1, S_2, S_i, \ldots, S_n\}$, and return encrypted speech data $[C]=\{[C_1],[C_2],[C_i],\ldots,[C_n]\}$.
5) Secure index constructing algorithm. $I \leftarrow SeIndex(pk,h_i)$. The algorithm inputs public key *pk* and deep binary hash codes $H=\{h_1, h_2, h_i, \ldots, h_n\}$, outputs secure index table I, and uploads it to CS storage.
6) Search trapdoor generation algorithm. $[T_q] \leftarrow TrapGen(G,pk)$. The algorithm inputs the public key *pk* and the deep binary hash code $G=(g_1, g_2, \ldots, g_k)$ of the querying speech W, and returns the secure index $[G]=([g_1],[g_2],\ldots,[g_k])$ as the search trapdoor $[T_q]$ of the DU.
7) Secure search algorithm. $[C_q] \leftarrow SearchEnc(evk,c,[T_q],I,[C])$. The algorithm takes the ciphertext calculation key *evk*, circuit c, search trapdoor $[T_q]$ corresponding to the querying speech W, secure index table I, and ciphertext $[C]$ as input, and CS executes security similarity calculation to return the search result $[C_q]$ corresponding to $[T_q]$.
8) Speech decryption algorithm. $S_q \leftarrow SpeechDec(sk,[C_q])$. The algorithm inputs the private key *sk*, encrypted speech $[C_q]$, and returns the decrypted speech S_q.

3.3 Secure speech retrieval scheme

Figure 2 is the processing flow chart of the secure speech retrieval method using CKKS FHE. The retrieval scheme mainly includes three parts: the construction of encrypted speech library (speech data encryption), the construction of triple deep hashing (speech feature extraction and secure index construction) and user retrieval.

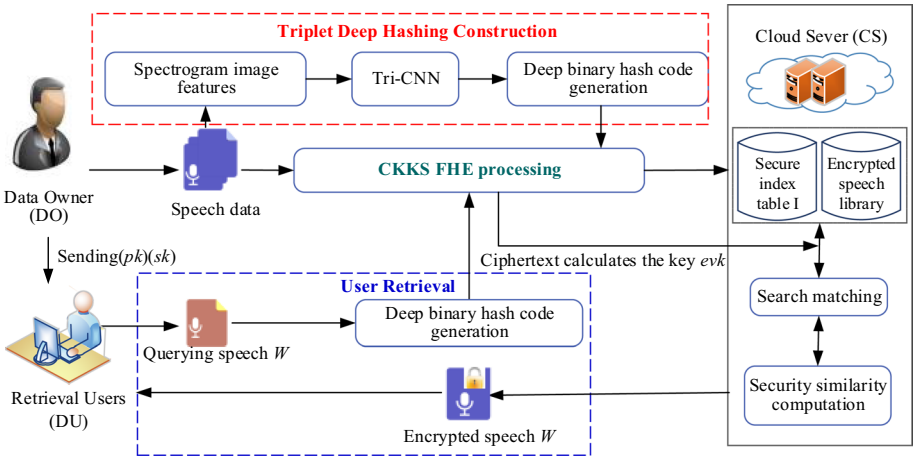


Fig. 2 Secure speech retrieval processing flow

1) Encrypted speech library construction (speech data encryption). DO preprocesses the local speech data sets $S=\{S_1, S_2, S_i, \ldots, S_n\}$ by framing, windowing, quantization, etc., and then uses the pk to perform CKKS FHE to obtain the encrypted speech data sets $[C]=\{[C_1], [C_2], [C_i], \ldots, [C_n]\}$, and uploads it to the encrypted speech library in CS for storage.
2) Triplet deep hashing construction. DO preprocesses the speech data sets S = {S1, S2, Si,…, Sn}, extracts the spectrogram image features, sends it to the Tri-CNN to extract the depth semantic feature vector, constructs the deep binary hash codes $H=\{h_1, h_2, h_i, \ldots, h_n\}$, and encrypts H with the same pk to obtain the secure index $[H]=\{[h_1], [h_2], [h_i], \ldots, [h_n]\}$, n is the number of speech data, and then sends $[H]=\{[h_1], [h_2], [h_i], \ldots, [h_n]\}$ to CS, One to one correspondence with encrypted speech data is stored in secure index table I.
3) Data user (DU). When retrieving speech data, DU preprocesses the querying speech W, extracts the spectrogram image features, and constructs the deep binary hash codes $G=(g_1, g_2, \ldots, g_k)$, and then use the same pk to encrypt G to obtain $[G]=([g_1], [g_2], \ldots, [g_k])$ (k is the length of the hash codes), and sends $[G]$ to CS as a search trapdoor $[T_q]$. When CS receives the retrieval request from DU, it searches and matches by cosine similarity, and returns the retrieval result to DU. When receiving the encrypted speech $[C_q]$ returned from the CS, DU decrypts it with the sk sent by the DO to obtain the plaintext speech data S_q.

3.3.1 Speech feature extraction and deep binary hash code generation

Figure 3 shows the proposed triple deep hash learning framework, which is used to generate deep binary hash codes with maximum discrimination and compactness.

The framework consists of three key components: 1) a novel triplet selection module, Group Hard, to mine an appropriate number of triplets for effective model training; 2) Tri-CNN for learning deep semantic information; 3) Triple loss function for pulling together anchor-negative spectrogram images and pushing away anchor-negative spectrogram images in hash space.

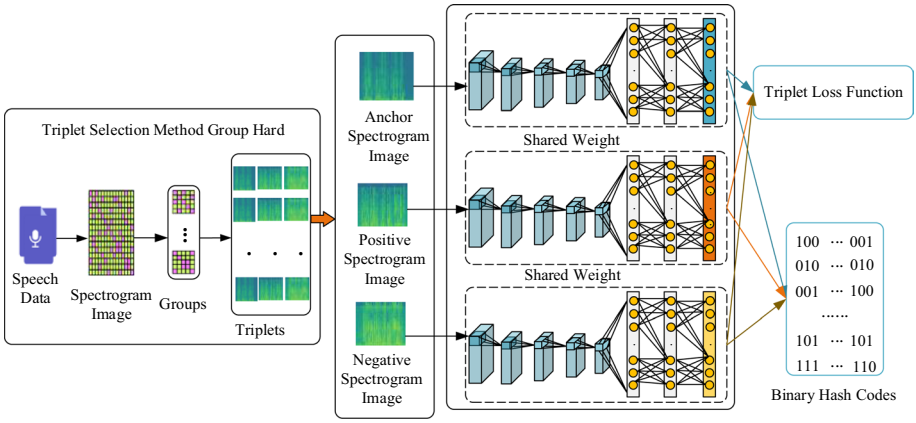


Fig. 3 Triplet deep hashing learning framework

As shown in Fig. 3, firstly, N spectrogram image features $\chi = \{x_i\}_{i=1}^{N}$ are given, in which the spectrogram image features x_i and x_j are given in pairs of similar labels s_{ij}, when x_i and x_j are similar, $s_{ij} = 1$ and $s_{ij} = 0$ when x_i and x_j are not similar; then the Tri-CNN is trained through the triplet spectrogram image feature group $\Gamma = \{t_i\}_{i=1}^{N_t}$, Each triplet $t_i = (x_i^a, x_i^p, x_i^n)$ is constructed from pairs of similar data, and the construction method is as follows:

For each anchor spectrogram image feature x_i^a, one positive spectrogram image feature x_i^p with $s_{ap} = 1$ (x_i^a and x_i^p are similar) and one negative spectrogram image feature x_i^n with $s_{an} = 0$ (x_i^a and x_i^n are not similar) can always be found.

For each triplet $t_i = (x_i^a, x_i^p, x_i^n)$, Tri-CNN maps the triplet t_i into a learned feature space with $f(t_i) =< z_i^a, z_i^p, z_i^n >$, z_i^a and z_i^p, z_i^n are the depth semantic feature vectors of anchor spectrogram image features x_i^a, positive spectrogram image features x_i^p and negative spectrogram image features x_i^n extracted by Tri-CNN, respectively, and then the deep binary hash codes is generated by learning the hash function.

When training the model, set the batch size of triplets to 128, and set the initial group number |Y| to 10 (THCHS-30 and TIMIT are used for the speech data sets). In the training process, it is necessary to ensure that $D(x_i^a, x_i^P) < D(x_i^a, x_i^n)$, where the similarity among x_i^a, x_i^p and x_i^n is measured by the Euclidean distance D between their deep features. Equation (1) is the triplet loss function:

$$L = \sum_{i=1}^{N_t} L_i = \sum_{i=1}^{N_t} \max\left(0, \delta - ||z_i^a - z_i^n||_2^2 + \delta - ||z_i^a - z_i^p||_2^2\right) \tag{1}$$

where δ is a threshold parameter between positive and negative pairs, and Γ is the set of cardinality N_t for all possible triplets in the training set.

Because Eq. (1) establishes a relative similarity relation between spectrogram image features, thus is much more reasonable than the absolute similarity relation used in previous pointwise or pairwise approaches.

1) Triplet selection module Group Hard.

Since there are a large number of triplet combinations in a data set, all training will seriously affect the convergence rate of the model. Therefore, an appropriate triplet

selection strategy can ensure a moderate number of triplets, which is crucial for effective model training and faster convergence.

The core idea of the triplet selection method in this paper is to randomly split the training data into several groups $\{Y_i\}_{i=1}^{|Y|}$, and then randomly select one hard negative spectrogram image feature for each anchor-positive pair spectrogram image features in one group. Triplets can be selected According to Eq. (2).

$$\Gamma = \bigcup_{i=1}^{|Y|} \bigcup_{a \in y_i} \bigcup_{p \in y_i^p} rand(y_i^n) \tag{2}$$

where $y_i^p = \{p \in y_i : p \neq a, s_{ap} = 1\}$ is the group of positive spectrogram image features similar to anchor spectrogram image feature a in the i-th group, $rand(y_i^n)$ is the random function that randomly chooses one negative spectrogram image feature from the group of hard negative spectrogram image feature $y_i^n = \{n \in y_i : \delta - ||z_i^a - z_i^n||_2^2 + \delta - ||z_i^a - z_i^p||_2^2, s_{an} = 0\}$. Here *hard* negative spectrogram image feature is defined as having non-zero loss value for a triplet $t_i = (x_i^a, x_i^p, x_i^n)$. Mining only the triplets with the hardest negative spectrogram image feature would select the outliers in the dataset and make it unable to learn the good relative similarity between the data. Thus, proposed triplet selection method in this paper based on the random sampling function $rand(y_i^n)$, and only select the negative examples with moderate hardness.

In the later period of model training, the average value of triplet loss will become smaller, and the number of hard triplet groups will become smaller. In order to ensure the effective training of the model, a decay strategy of the number of groups $|Y|$ is designed: if the actual number of valid hard triplets is lower than the minimum number of the valid hard triplet, the size of the groups is halved until $|Y| = 1$.

Complexity analysis: Because the triplet selection method in this paper can remove invalid triplets with zero loss ($L_i = 0$), resulting a valid triplet set Γ whose size $|\Gamma|$ is much smaller than the possible number N^3 of triplets. Through proposed selection strategy that chooses one negative spectrogram image feature for each anchor-positive spectrogram image feature, the number of the candidate triplets for training is further reduced to $|\Gamma|/|Y|$. Furthermore, all the selected triplets are hard triplets ($L_i > 0$, Eq. (1)), and the total amount can be controlled in a suitable range by adjusting the number of groups Y, thus realizing effective triplet training and generating efficient and compact hash codes.

2) Learning part of spectrogram image features.

Tri-CNN is used to extract the feature representation suitable for binary hash code learning. The configuration of each CNN network model is shown in Table 1.

Tri-CNN includes three CNN network models sharing weights, each CNN network model includes five convolution layers and three fully connected layers, and the last fully connected layer is replaced by a fully connected layer, and the output is mapped as a binary hash code. The convolution layer includes Filter, Filter size, Stride, Padding, Pooling, etc.

3) Triplet loss function.

The triplet deep hash method is a learning process that uses the triplet loss function for pulling together deep binary hash codes of anchor spectrogram image features and positive spectrogram image features (similar), and pushing away deep binary hash codes of the negative spectrogram image features (dissimilar). The process is shown in Fig. 4.

4) Structure deep binary hash codes.

The existing deep hashing methods have some limitations. If the gradient-based method is used to directly optimize the objective function, it will be constrained by

Table 1 Configuration table of model parameters

Layer	Setting Parameters				
	Filter	Filter size	Stride	Padding	Pooling
Convolution layer 1	64	11×11	4×4	2×2	3×3
Convolution layer 2	192	5×5	1×1	2×2	3×3
Convolution layer 3	384	3×3	1×1	1×1	-
Convolution layer 4	256	3×3	1×1	1×1	-
Convolution layer 5	256	3×3	1×1	1×1	3×3
Fully connected layer	9216				
Fully connected layer	4096				
Hash layer (FC)	Deep binary hash code				

discrete values, resulting in low retrieval accuracy of the generated deep binary hash codes. In order to solve this problem, the depth hash code H is first relaxed to be continuous, and binarization is no longer required, that is, the output value is first limited to the range of (-1,1) by using the *tanh* function, and the continuous value is regularized to approximate the required discrete value (+ 1/-1). Then the desired discrete depth hash codes $H \in \{-1, 1\}^k$ is output using Eq. (3).

$$H = u(X) \in \{-1, 1\}^k \tag{3}$$

where X is the deep semantic feature extracted by Tri-CNN, and $u(\cdot)$ is the parameter value of convolution layer, pool layer and full connection layer of Tri-CNN.

Finally, the Tri-CNN maps the discrete depth hash code into binary hash codes with length k by the designed symbolic function $sign(\cdot)$ with Eq. (4).

$$h_{i,k} = sign(H - S_{meadian}) \tag{4}$$

Combining Eq. (3) and Eq. (4), the generation process of deep binary hash codes can be shown in Eq. (5).

$$h_{i,k} = sign(H - S_{meadian}) = \begin{cases} 1, H \geq S_{meadian} \\ 0, otherwise \end{cases} \tag{5}$$

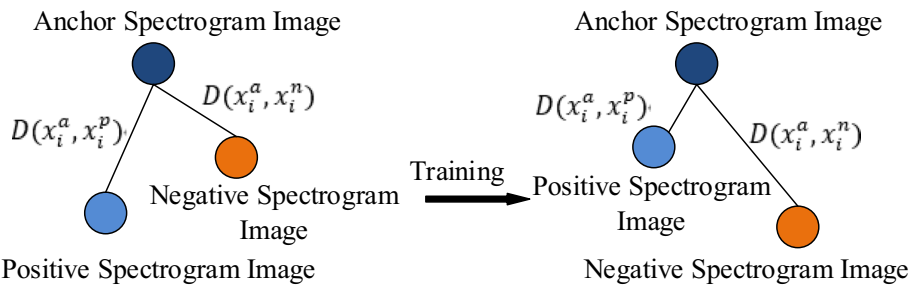


Fig. 4 The function of triplet loss function

where $S_{meadian}$ represents the median of the semantic feature vector values of the feature deep of each spectrogram image in [0,1] interval.

3.3.2 Construction of encrypted speech library and secure index

In order to ensure the privacy and security of cloud speech data and deep binary hash codes, this paper designs a speech CKKS FHE scheme to encrypt it, constructs an encrypted speech library and secure index table I, and uploads them to CS for storage. Figure 5 shows a flowchart of speech encryption processing.

1) SpeechEnc(pk, S): The specific processing of DO encrypted original speech is as follows:

Step 1: Analog-to-digital conversion (A/D). Read the original speech signal for sampling quantization processing, and then digitize the processed analog signal, speech data in the complex vector space $C^{N/2}$ (N is generally a power of 2).
Step 2: Preprocessing. Preemphasis, framing and windowing of digital signals.
Step 3: Rescaling. Step 3: Take element $z \in C^{N/2}$ and expand it to obtain $\pi^{-1}(z) \in H$ (π is a natural projection map, H is a subring of $C^{N/2}$); To ensure the calculation accuracy, multiply by the scaling factor Δ and enlarge the integer part.
Step 4: Segmentation. In order to reduce the computational complexity and increase the amount of communication, the real vector value representing the speech data is segmented according to the polynomial module.
Table 2 shows the encryption parameters of speech encryption using CKKS FHE scheme.

Step 5: Batch packaging. Using CRT and SIMD batch processing technology, the real value after segmentation is packed onto a series of matrices respectively for parallel operation.
Step 6: polynomial cyclic coding. To ensure the subsequent accuracy, multiply all the matrices packaged in batch by the scaling factor Δ and the reciprocal of the gauge embedding map σ, so that $\sigma^{-1}(\Delta \cdot \pi^{-1}(z)) \in R[X]/(X^N + 1)$, where $R[X]/(X^N + 1)$ is a polynomial quotient ring isomorphic to $C^{N/2}$, namely, the encoding process is $m(X) = \sigma^{-1}(\Delta \cdot \pi^{-1}(z)) \in R[X]/(X^N + 1)$.
Step 7: Rounding polynomial coefficients. In order to ensure that the decoded vector value is close enough to the original vector value, coordinate-wise random rounding

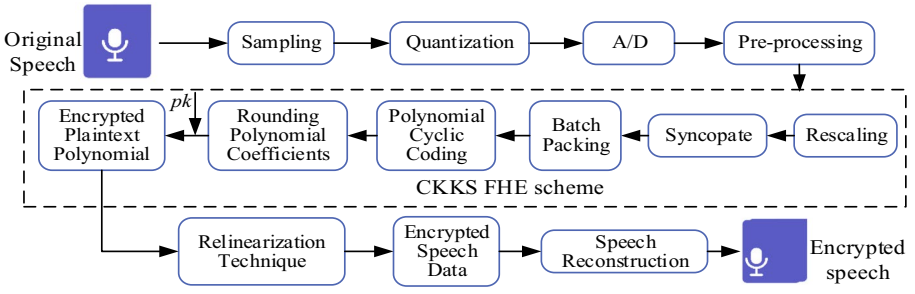


Fig. 5 Speech Encryption Processing Flow Chart

Table 2 Encryption Parameters of Speech Encryption

Name	Parameter
Encryption	CKKS
Polynomial module	8192
Ciphertext module	218bits
Scaling factor	41

method is adopted, and the polynomial coefficients are rounded down and up according to the distribution probability of the distance from adjacent integers, that is, the rounding process can be expressed as $m(X) = \lfloor \sigma^{-1}(\Delta \cdot \pi^{-1}(z)) \rceil \in Z[X]/(X^N + 1)$.

Step 8: Encrypt the plaintext polynomial. Generate $r \leftarrow X_r$ and $e_0, e_1 \leftarrow X_e$, and for plaintext polynomial $m(X) = \lfloor \sigma^{-1}(\Delta \cdot \pi^{-1}(z)) \rceil \in Z[X]/(X^N + 1)$, output ciphertext $[C] \leftarrow r \cdot pk + (m(X) + e_0, e_1) \bmod Q$.

Step 9: Relinearization technique. Relinearize the ciphertext $[C] \leftarrow r \cdot pk + (m(X) + e_0, e_1) \bmod Q$, reduce the dimension of the ciphertext, and ensure the dimension of the ciphertext.

2) Index(pk, h_i): The specific processing of DO encryption deep binary hash code is as follows:

Table 3 shows the encryption parameters of constructing secure index by CKKS FHE scheme.

Step 1: Extract the deep hash codes sets. The trained Tri-CNN is used to extract the deep binary hash codes $H=\{h_1, h_2, h_i, \ldots, h_n\}$ of the original speech data set $S=\{S_1, S_2, S_i, \ldots, S_n\}$;

Step 2: Plaintext coding. The deep binary hash code corresponding to each speech data is regarded as a plaintext vector, and then all binary hash codes $H=\{h_1, h_2, h_i, \ldots, h_n\}$ are coded with plaintext polynomial by using the Encode coding function in SEAL library to realize the subsequent encryption operation.

Step 3: Encrypt the plaintext polynomial. Encrypt the plaintext polynomial with the same public key pk combined with the Encrypted encryption function to obtain the secure indexes $[H]=\{[h_1], [h_2], [h_i], \ldots, [h_n]\}$ of all speech data. After the encryption is completed, DO uploads the secure index $[H]$ corresponding to the encrypted speech data $[C]$ to the secure index table I for storage.

3.4 User retrieval

Data user (DU) Perform the following steps to query speech files on the client:

Table 3 Encryption parameters for constructing a secure index

Name	Parameter
Encryption	CKKS
Polynomial module	8192
Ciphertext module	218bits
Scaling factor	34
Iterations	1
Multiplication levels	4

Step 1: Extract the deep binary hash codes. The trained Tri-CNN is used to extract the deep binary hash code $G=(g_1, g_2, \ldots, g_k)$ of the querying speech W;
Step 2: Plaintext encoding. The deep binary hash code $G=(g_1, g_2, \ldots, g_k)$ corresponding to the querying speech W is regarded as a plaintext vector, and the Encode function in SEAL library is used for plaintext polynomial encoding to realize the subsequent encryption operation;
Step 3: Encrypt the plaintext polynomial. Encrypt the plaintext polynomial with the same public key pk combined with the Encrypted encryption function, using the search trapdoor generation algorithm in Section 3.2 to obtain the secure index $[G]=([g_1], [g_2], \ldots, [g_k])$. After encryption, DU sends the secure index $[G]=([g_1], [g_2], \ldots, [g_k])$ to the CS as the search trapdoor $[T_q]$ of the querying speech W.
Step 4: Retrieve data. When CS receives the search trapdoor $[T_q]$ sent by the searching user, it performs security similarity calculation by using cosine distance and secure indexes $[H]=\{[h_1], [h_2], [h_i], \ldots, [h_n]\}$ in the secure index table I. Equation (6) is the cosine similarity formula:

$$D = \frac{< [h_i], [T_q] >}{||[h_i]||||[T_q]||} \geq / < \theta \tag{6}$$

where $[h_i], [T_q] \in \{0,1\}^{k\times 1}$, k represent the length of the deep binary hash code.
To reduce the noise of the channel, a projection matrix such as Eq. (7) is used:

$$\begin{aligned} D &= \frac{(P^T[h_i])^T(P^T[T_q])}{\sqrt{(P^T[h_i])^T(P^T[h_i])}\sqrt{(P^T[T_q])^T(P^T[T_q])}} \\ &= \frac{[h_i]^T M[T_q]}{\sqrt{[h_i]^T M[h_i]\cdot[T_q]^T M[T_q]}} \end{aligned} \tag{7}$$

where $M = PP^T \in \{0,1\}^{k\times k}$, the matrix M is obtained by the server in the training process, and it is not from speech data.
Although FHE supports addition and multiplication in encrypted domain, it is not feasible to find the reciprocal or square root of a number. However, the cosine similarity in Eq. (7) needs to calculate the inverse square root operation, so using Newton–Raphson method, only multiplication and addition can be used instead of this operation.
Define the function $f(x) = \frac{1}{x^2} - a$, where the root of this function is the inverse square root function of a, that is $\frac{1}{x^2} - a = 0$, and obtain $x = \frac{1}{\sqrt{a}}$; The root of Newton–Raphson method can be solved according to Eq. (8):

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{8}$$

where the reciprocal of $f(x)$ in $x = x_n$ is $f'(x_n) = -\frac{2}{x_n^3}$. Equation (8) can be revised to Eq. (9):

$$x_{n+1} = 1.5x_n - 0.5ax_n^3 \tag{9}$$

In order to find the inverse of the square root of a, the Eq. (9) must be repeatedly calculated. The required number of iterations depends heavily on x_0, which is the initial value of Eq. (9). If $a_{\min} < a < a_{\max}$, then the best value of a is the average value of the bound-

ary, namely $x_0 = \frac{1}{\sqrt{a_{\min}}} + \frac{1}{\sqrt{a_{\max}}}$. With the initialization x_0, Eq. (9) can be calculated only by multiplication, so it friendly replaces the inverse square root operation.
By combining Eq. (7) and (9), Eq. (10) and (11) can be obtained:

$$D = \frac{1}{\sqrt{[a]}} \cdot [h_i]^T M[T_q] \tag{10}$$

$$[a] = [h_i]^T M[h_i] \cdot [T_q]^T M[T_q] \tag{11}$$

Since $[a]$ in Eq. (11) is encrypted, the inverse of the square root of $[a]$ in Eq. (10) cannot be directly calculated. Therefore, using Newton–Raphson method, the approximate result after the first iteration according to Eq. (9) is Eq. (12):

$$\frac{1}{\sqrt{[a]}} \approx_1 1.5x_0 - 0.5[a]x_0^3 \tag{12}$$

The result of the second iteration is Eq. (13):

$$\frac{1}{\sqrt{[a]}} \approx_2 1.5(1.5x_0 - 0.5[a]x_0^3) - 0.5[a](1.5x_0 - 0.5[a]x_0^3)^3 \tag{13}$$

And so on and so forth ($\approx_1$ and $\approx_2$ denote approximation after first and second iterations, respectively). Using the first approximate value in Eq. (12), the approximate value of Eq. (10) can be obtained as shown in Eq. (14):

$$D \approx_1 (1.5x_0 - 0.5[a]x_0^3)[h_i]^T M[T_q] \tag{14}$$

According to Eq. (10) and (11), Eq. (14) can be expanded into Eq. (15):

$$\begin{aligned} D &\approx_1 1.5x_0[^{h_i}M[T_q] - 0.5x_0^3[^{h_i}M[h_i] \cdot [^{T_q}M[T_q][^{h_i}M[T_q] \\ &\approx_1 [^{1.5}M[T_q] - [^{0.5}M[h_i] \cdot [^{T_q}M[T_q][^{h_i}M[T_q] \\ &\approx_1 \underbrace{\underbrace{[1.5x_0h_i]^T M[T_q]}_{Level1}}_{Level2} - \underbrace{\underbrace{[0.5x_0^3h_i]^T M[h_i]}_{Level1} \cdot \underbrace{[T_q]^T M[T_q]}_{Level1}}_{Level2} \underbrace{[h_i]^T M[T_q]}_{Level1} \\ &\qquad \underbrace{\qquad\qquad\qquad\qquad\qquad\qquad}_{Level2} \qquad\qquad\qquad\qquad \underbrace{\qquad\qquad\qquad\qquad}_{Level2} \qquad\qquad \underbrace{\qquad\qquad}_{Level2} \\ &\qquad\qquad\qquad\qquad\qquad\qquad

1) Restore speech data. Using the private key *sk* to decode the ciphertext polynomial into a plaintext polynomial, and then multiplied by Δ^{-1} to restore the polynomial coefficients of real number type.
2) Inverse operation. Through the inverse process of polynomial coding, $\pi(\sigma(\Delta^{-1}m(X))) \in C^{N/2}$ is calculated, and all real number types of speech data are obtained.
3) Restore speech. Set the appropriate sampling rate, sampling points, quantization numbers, etc., and recover the speech data into a speech file to get the decrypted speech.

4 Performance comparison and experimental simulation

In order to evaluate the performance of the proposed method, the Chinese speech database THCHS-30 [29] released by the Center for Language and Language Technology (CSLT) of Tsinghua University and TIMIT [30] released by Texas Instruments, Massachusetts Institute of Technology and Stanford Research Institute were respectively used as data sets, with a sampling frequency of 16 kHz and a sampling accuracy of 16bit of single-channel wav speech s. There are 1,000 news clips with different speech contents in THCHS-30, totaling 13,388, each about 10 s in length, with a total length of about 30 h. The TIMIT data set consists of 630 people from eight major dialect areas in the United States, each speaking a given 10 speech clips, totaling 6,300, each about 4 s in length.

In the experiment of this paper, 10 speech clips with different contents were selected from THCHS-30 and TIMIT respectively, and various content preserving operations such as volume adjustment, requantization, noise addition, MP3 and resampling were carried out to obtain a total of 10,000 and 11,000 speeches, with 10 categories. The former had 1,000 speech clips per category, while the latter had 1,100, where 80% were used for training and 20% for testing. In the experimental analysis stage, 1,000 and 10,000 pieces of speech data were randomly selected from the original speech library to evaluate recall rate, precision rate, F1 score and retrieval time, respectively.

The experimental hardware platform is: cloud server RTX 3060, memory 32.1 GB, disk 451.0G, 12.6G video memory. Software environment: Windows 10, PyCharm 2021.1.1, MATLABR2017b software simulation, deep learning framework PyTorch v1.10.

4.1 Triplet selection module Group Hard performance analysis

In this paper, the triplet selection strategy Group Hard adopted can randomly extract a certain amount of effective triplets with hard samples from several randomly divided spectrogram image feature groups, which makes the training effective, the convergence speed faster, and the retrieval performance more accurate. Table 4 shows a MAP of different bits on TIMIT.

As can be seen from Table 4, the online triplet selection method has the lowest performance, because the online triplet selection method directly selects all hard triplets from the ungrouped spectrogram image features for each anchor spectrogram image feature group, and the proportion of effective hard triplets for training is low. The offline triplet selection method performs slightly worse because it generates too many candidate triplets in model training and wastes most of the training batches. The triplet selection method presented in this paper has the best performance because it reduces the obsoletion effect of the offline triplet selection method, dividing the training data into

Table 4 Shows a MAP of different bits on TIMIT (%)

Method	64	128	256	384
Online triplet selection	71.21	84.86	91.35	91.39
Offline triplet selection	89.67	93.56	93.61	93.68
Group Hard	93.69	95.62	95.67	95.74

Table 5 Shows proportion of non-obsolete hard triplets (amount)

Method	0	1000	2000	3000	4000
Offline triplet selection	55.91	21.38	15.26	13.20	5.99
Group Hard	85.01	50.68	22.67	15.59	13.66

specific groups, selecting hard triplets in each group, and reduceing the number of training triplets from Γ to $|\Gamma|/|Y|$.

Table 5 shows proportion of non-obsolete hard triplets.

As can be seen from Table 5, when the training data is divided into a specific Group $|Y|$, the number of outdated hard triples is significantly reduced, which verifies the validity of the offline selection strategy Group hard in this paper. Therefore, the triplet selection method in this paper adopts the offline selection strategy Group hard.

4.2 Performance analysis of Tri-CNN

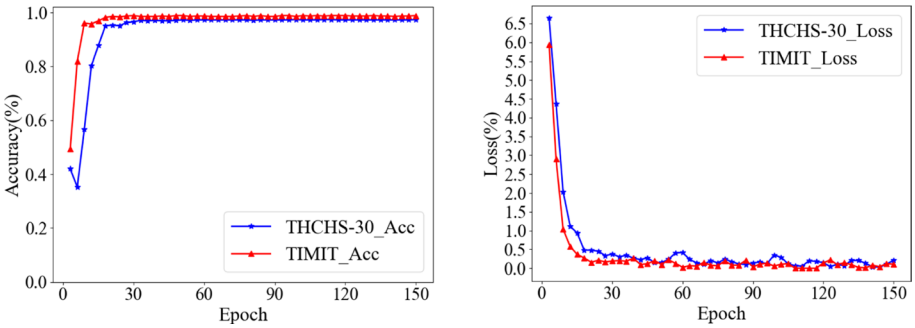
Because of its powerful data learning ability, CNN is widely used in the field of image retrieval to extract the deep semantic features of images. Therefore, in this paper, Tri-CNN sharing three weights is used to extract the deep semantic features of spectrogram image features, and the performance of the model is evaluated by pre-training the model. Figure 6 shows the test accuracy and loss curve of Tri-CNN.

As can be seen from Fig. 6, with the speech data in THCHS-30 and TIMIT as the input of the Tri-CNN, the convergence speed is fast and the accuracy is almost close to 1. After 150 iterations, the test accuracy is 99.96% and the test loss is gradually close to 0. In contrast, when the speech data in TIMIT is used as input, the convergence is faster, the accuracy is higher, and the loss is lower after 150 iterations. Therefore, the network model used by the method in this paper has a good degree of fitting to the input data, without overfitting or underfitting phenomenon, and has superior performance under both data sets.

4.3 Comparison of classification accuracy and computational efficiency of different security similarity calculation

In speech security retrieval, cosine similarity, Euclidean distance [28] and Pearson correlation coefficient [28] are often used to calculate the distance between two samples. The Euclidean distance calculation process as shown Eq. (16).

$$D_1 = \sum_{i=1}^{n} \left([\widetilde{h}_i] - [(\widetilde{T}_q)_i]\right)^2 \tag{16}$$



(a) Accuracy curve (b) Loss curve

Fig. 6 Test accuracy and loss curve of Tri-CNN

where $[\widetilde{h}_i]$ and $[\widetilde{T}_q]$ represent standardized data.

Since the size of different feature indexes of secure indexes has a great influence on Euclidean distance, Pearson's correlation coefficient, which is insensitive to size, is often chosen to measure the similarity between two samples in some applications. The calculation process as shown Eq. (17).

$$D_2 = \frac{([h_i]-[\overline{h}_i])^T([T_q]-[\overline{T}_q])}{\|[h_i]-[\overline{h}_i]\|\|[T_q]-[\overline{T}_q]\|} = \frac{[h_i]^T[\widetilde{T}_q]}{n-1} \tag{17}$$

where Eq. (17) has standardized the data before calculation, $[\overline{h}_i]$ represents the mean value of each secure index in secure index table I, $[\overline{T}_q]$ represents the mean value of search trapdoor, and n represents the number of encrypted speech data in the encrypted speech library.

In order to evaluate the classification accuracy and calculation efficiency of the proposed method under different security similarity measurement methods, 1,000 and 10,000 speech clips are randomly selected, and they are used as the user's querying speech clips after MP3 compressed content preserving operation, and the trapdoor generation algorithm of DU in Section 3.4 is used to generate the search trapdoor of the querying speech clips. When retrieving and matching, the similarity threshold T is set to 0.234. If the distance $d([H],[T_q]) < T$ between the search trapdoor of the querying speech clips and the secure index in the cloud secure index table I, the retrieval is successful. Figure 7 shows the comparison results of cosine similarity, Euclidean distance, Pearson correlation coefficient classification accuracy and calculation efficiency. The experimental data in Fig. 7 is the mean value of eight experimental results.

It can be seen from Fig. 7(a) and Fig. 7(b) that the classification accuracy of Euclidean distance and cosine similarity is stable at about 97%, and the classification accuracy of Pearson correlation coefficient is as low as about 40%. However, compared with the Euclidean distance, the classification accuracy of cosine similarity is more stable. It can be seen from Fig. 7(c) and Fig. 7(d) that with the increase of the number of querying speech data clips in the test set, the time consumption of the three similarity measurement methods increases linearly. In contrast, the calculation efficiency of Euclidean distance is about 0.05 s faster than that of cosine similarity, because the latter requires more ciphertext multiplication operations. Therefore, under the premise of comprehensive consideration of

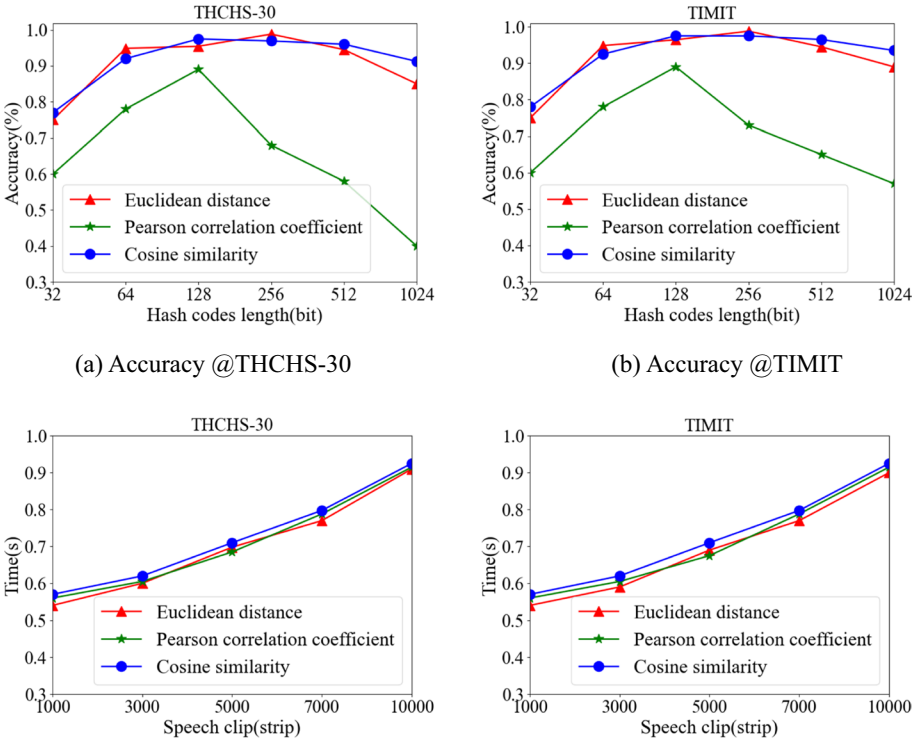
classification accuracy and calculation efficiency, cosine similarity is more suitable as a measurement method for massive speech security retrieval.

4.4 Retrieval performance evaluation

4.4.1 Retrieval accuracy

In order to get the node settings of the optimal network model, the mean Average Precision (mAP) value can be selected to evaluate the retrieval accuracy of the adopted network model under different hash codes lengths. In the experiment, the binary hash codes and secure index (encrypted binary codes) of the speech data after the content preserving operation are used for testing. Firstly, the AP value of the speech is calculated according to Eq. (18), and then the mAP value is calculated according to Eq. (19).

$$AP(x_q) == \frac{1}{R_k}\sum_{k=1}^{n} p(k)\Delta r(k) \tag{18}$$



(a) Accuracy @THCHS-30

(b) Accuracy @TIMIT

(c) Efficiency @THCHS-30

(d) Efficiency @TIMIT

Fig. 7 Comparison of classification accuracy and calculation efficiency

$$mAP = \frac{1}{Q}\sum_{q=1}^{Q} AP(x_q) \tag{19}$$

where R_k is the number of relevant speech clips, $p(k)$ is the precision at cutoff k in the returned list, and $\Delta r(k)$ is an indicator function which equals 1 if the k-th returned speech clip is a ground-truth neighbor of x_q. Otherwise, $\Delta r(k)$ is 0. Given Q queries, mAP is the AP of all retrieval results sorted.

Table 6 shows the mAP values of Tri-CNN in plaintext domain and encrypted domain under different data sets and different hash code lengths.

As can be seen from Table 6, the mAP value of Tri-CNN in the THCHS-30 and TIMIT data sets in the plaintext domain increases with the growth of the hash code length, because the longer the hash code, the richer the semantic information. When the hash code length is 128bit, the mAP value of the Tri-CNN has reached more than 95%. In addition, the mAP values of 128bit and 256bit have little difference, indicating that when the number of hash layer nodes of the Tri-CNN is 128bit, the model has reached the optimal performance. Compared with the plaintext domain, when the hash code length is 128bit, the retrieval precision in the encrypted domain decreases the least, and the optimum is obtained. Therefore, proposed method sets the number of nodes at the hash layer of the final network model as 128bit, and then obtains the binary hash code of 128bit for subsequent retrieval and analysis.

4.4.2 Analysis of recall rate, precision rate and F1 score

In this paper, the Recall rate, Precision rate and F1 score were used as evaluation criteria to measure the performance of the retrieval system. Recall rate (R) represents the proportion of queried samples to queried related samples in the list returned by retrieval. The calculation formula as shown Eq. (20).

$$R = \frac{TP}{TP + FN} \times 100\% \tag{20}$$

where TP and FN respectively represent the number of retrieved speech clips related to the query, and the number of speech clips related to the query are not retrieved, and the sum of TP and FN is the total number of speech clips related to the query in the speech library.

Precision rate (P) is the retrieval accuracy, which represents the proportion of speech segments whose query is true in the returned list. The calculation formula as shown Eq. (21).

$$P = \frac{TP}{TP + FP} \times 100\% \tag{21}$$

Table 6 Comparison of mAP with different data sets and different hash code lengths (%)

Hash code length (bit)	THCHS-30		TIMIT	
	Plaintext	Ciphertext	Plaintext	Ciphertext
64	93.67	90.90	93.69	90.97
128	95.56	93.71	95.62	93.73
256	95.61	93.69	95.67	93.70
384	95.68	93.62	95.74	93.69

where *FP* represents the total number of retrieved speech clips unrelated to the query, and the sum of *TP* and *FP* is the total number of returned speech clips.

Because recall rate and precision rate are anti-dependent, F1 score is used as the test index. The higher the score of F1, the better the retrieval performance. The calculation formula as shown Eq. (22).

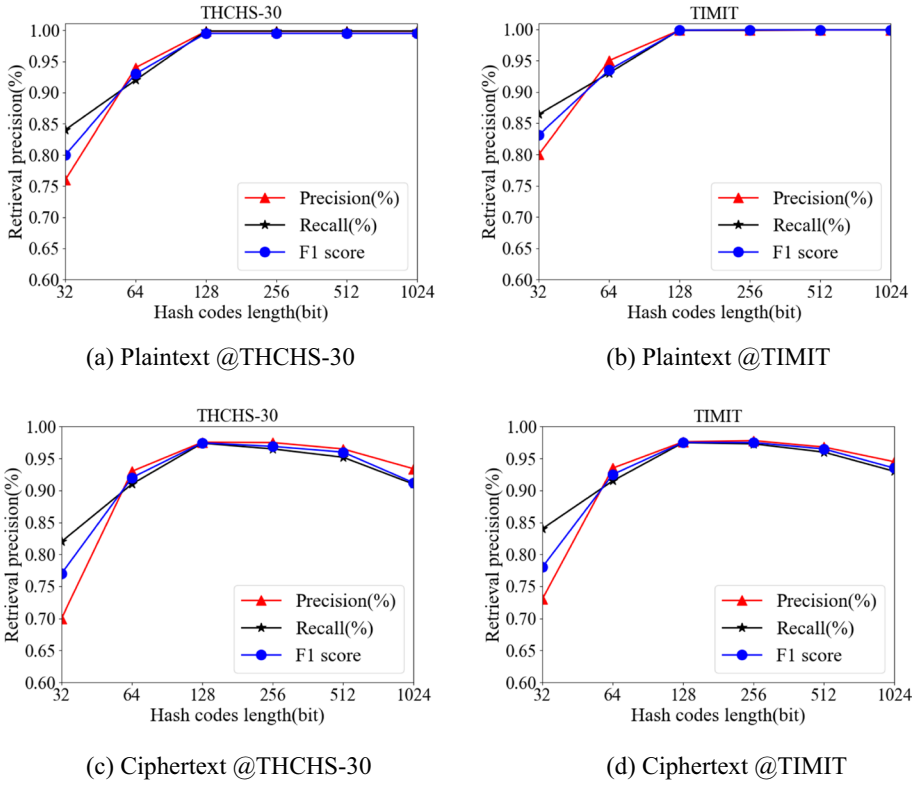
$$F_1 = 2 \times \frac{P \times R}{P + R} \tag{22}$$

In order to verify the retrieval performance of proposed method in plaintext domain and encrypted domain, the deep binary hash codes with different lengths are used to evaluate the recall rate, precision rate and F1 score. Figure 8 shows the comparison of recall rate, precision rate and F1 scores of binary hash codes with different lengths and depths under THCHS-30 and TIMIT data sets in plaintext domain and encrypted domain. Figure 8(a) and Fig. 8(b) show the comparison of THCHS-30 and TIMIT data sets in the plaintext domain respectively; Fig. 8(c) and Fig. 8(d) show the comparison of THCHS-30 and TIMIT data sets in the encrypted domain respectively.

As can be seen from Fig. 8(a) and 8(b), in plaintext domain, the retrieval performance of the proposed method under THCHS-30 and TIMIT data sets is continuously improved with the increasing length of the deep binary hash codes. When the deep binary hash code becomes 128bit, its retrieval performance has become flat, which is equivalent to the retrieval performance of 256, 512 and 1,024bit deep binary hash codes, indicating that the 128bit deep binary hash codes can meet the basic requirements of the retrieval system. It can be seen from Fig. 8(c) and Fig. 8(d) that when the deep binary hash codes of proposed method is 128 bits, the retrieval performance of THCHS-30 and TIMIT data sets has become flat, but the recall rate, precision rate and F1 score in plaintext domain decrease by about 2%. When the deep binary hash codes is greater than 512bit, its value begins to have a significant downward trend. This is because the encryption scheme in this paper encrypts the depth binary hash codes, increasing the computational complexity of polynomial module, ciphertext module, multiplication depth, iteration number, etc., resulting in partial information loss. When the length of the deep binary hash codes is greater than 512bit, its computational complexity exceeds the critical value that the retrieval system can accept, so the recall rate, precision rate and F1 scores begin to decline. Therefore, the proposed method adopts 128 bit deep binary hash codes, which can reduce the matching length and obtain good recall rate, precision rate and F1 score.

To better illustrate the discrimination ability of the deep binary hash code generated by proposed method, T-SNE (T-Distributed Historical Neighbor Embedding) [31] is used to visualize it. Because the encryption scheme in this paper is HE, which only encrypts the semantic information of data and supports ciphertext operation, the encrypted data still retains the relative similarity of the original plaintext data, and the function of t-SNE is to transform the "relative similarity of data" into a probability distribution for visualization. Therefore, the visualization effect in plaintext domain is the same as that in encrypted domain. Take the visualization of the distribution of 128 bit deep binary hash codes in THCHS-30 and TIMIT data sets in plaintext domain as an example. Figure 9 shows the distribution visualization of deep binary hash codes in different data sets, Fig. 9(a) shows the visualization in THCHS-30 data set, and Fig. 9(b) shows the visualization in TIMIT data set.

It can be observed from Fig. 9 that the deep binary hash codes of proposed method in the THCHS-30 and TIMIT data sets show a clear distinguishing structure, and different



(a) Plaintext @THCHS-30

(b) Plaintext @TIMIT

(c) Ciphertext @THCHS-30

(d) Ciphertext @TIMIT

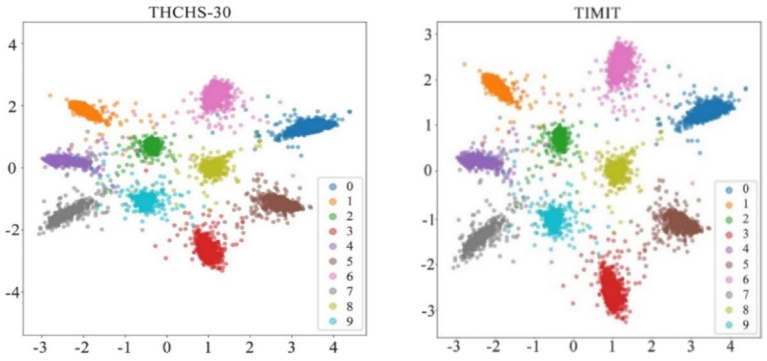
Fig. 8 Comparison of recall rate, precision rate and F1 scores under different data sets

types of data points are well separated, which shows that proposed method effectively promotes the training of Tri-CNN, fully extracts the semantic features of spectrogram images, and the triplet loss function adopted by the model fully optimizes the depth binary hash codes, making them have larger out-of-class differences and smaller in-class differences. Therefore, the deep binary hash codes generated by proposed method has stronger discrimination and can achieve more accurate speech secure retrieval.

4.4.3 Performance comparison with existing speech retrieval methods

In order to objectively analyze the retrieval performance of the proposed method, this paper uses 128 bit deep binary hash codes to compare the performance with the existing speech retrieval methods [20–22, 32–34]. Table 7 shows the performance comparison between proposed method and other existing retrieval methods.

It can be seen from Table 7 that the recall rate (R), precision rate (P), F1 score and retrieval efficiency of the proposed method are better than those in Ref. [21, 22], which shows that proposed method has good universality for speech data, generates efficient and compact binary hash codes, and obtains good retrieval performance. In terms of retrieval means, Ref. [21, 22] all used AES encryption algorithm to construct encrypted speech library, encrypted MFCC feature vector with fuzzy matrix to generate keywords, and store the encrypted speech library and keywords in the cloud in one-to-one correspondence. In the retrieval process, the matching retrieval



(a) THCHS-30 (b) TIMIT

Fig. 9 Visualization of the distribution of deep binary hash codes in different data sets

of encrypted domain is realized by using keywords. Ref. [20] has constructed the encrypted speech library and secure index table by pseudo-random function, Ref. [32] has constructed the encrypted speech library and secure index table by hyperchaotic Lorentz compressed sensing encryption algorithm, and Ref. [20, 32] have realized the matching search of encrypted domain by its secure index. Ref. [33, 34] respectively use the secure hash algorithm (SHA256) and the chaotic encryption system based on Rossler mapping to construct the encrypted speech library, while the binary hash codes do not use any secure means, which actually completes the matching search in the plaintext domain. In this paper, CKKS FHE scheme is adopted to construct the encrypted speech library and secure index, which realizes the matching search of encrypted domain and better realizes the privacy protection and secure sharing of speech data. The recall rate, precision rate, F1 score and retrieval efficiency of proposed method are higher than those of Ref. [21, 22], which is mainly because Ref. [21, 22] only uses the primary speech feature MFCC for retrieval matching, while proposed method hash constructs the deep speech features extracted by Tri-CNN, and generates low-dimensional binary hash codes rich in semantic information, thus effectively improving the retrieval performance. The retrieval efficiency of proposed method is obviously better than that of Ref. [21, 22], because the total number of querying speech in this paper is 9,700 more than that in Ref. [21, 22], and the time difference is only about 0.7 s. The recall rate, precision rate and F1 score of proposed method are lower than those of Ref. [32, 34], which is because Ref. [32, 34] generates binary hash code with longer bits, and Ref. [32] increases the total number of frames by increasing the number of overlapping frames, thus enriching the semantic information. However, the calculation of spectrogram image features used in this paper is too large, resulting in the loss of some semantic information. The retrieval efficiency of proposed method is lower than that of Ref. [32, 34], on the one hand, because the total number of querying speech in Ref [32, 34] is only 1,000, on the other hand, Ref [32, 34] uses a second search, that is, Ref [32] first classifies the extracted spectral centroid by KNN, and then binarizes the feature vector to generate a one-way hash, which only one item needs to be matched in the intra class matching process, resulting in the retrieval range and the number of matches. Ref [34]. first classified it according to linear prediction minimum mean square error and inverse fast Fourier transform (LPM-IFFT), and then performed binary differential hash processing, and the retrieval process only needs to match the feature index related to the speech in the feature index table. The retrieval process only needed to match the feature index related to the speech in the feature index table. However, the total number of querying speech in proposed

Table 7 Comparison of retrieval performance with existing speech retrieval methods

Methods	Network model	Data set	Speech feature	Means of retrieval	*P* (%)	*R* (%)	F1 (%)	mAP (%)	Total of speech clips	Retrieval time (s)
Ref. [20]	LSTM	THCHS-30	MFCC	Ciphertext	100	100	100	-	1,000	0.2690
			Spectrogram		74.97	92.03	82.63	-		0.2740
Ref. [21]	Nothing	Homemade corpus	MFCC	Ciphertext	95.00	93.00	-	-	300	0.170
Ref. [22]	Nothing	Homemade corpus	MFCC	Ciphertext	94.00	93.00	93.50	-	300	0.139
Ref. [32]	Nothing	TIMIT	Centroid spectrum	Ciphertext	100	100	-	-	1,000	0.0017
Ref. [33]	CNN	THCHS-30	Spectrogram	Plaintext	100	100	-	96.31	10,000	0.4854
Ref. [34]	Nothing	TIMIT	LPM-IFFT	Plaintext	100	100	-	-	1,000	0.0895
Proposed	Tri-CNN	THCHS-30	Spectrogram image	Plaintext	99.85	99.88	99.86	95.56	10,000	0.2831
				Ciphertext	97.40	97.55	97.46	93.71		0.9073
		TIMIT		Plaintext	99.90	99.92	99.91	95.62		0.2765
				Ciphertext	97.48	97.61	97.52	93.73		0.9068

method is 10 times that in Ref. [32, 34], and the exhaustive search is adopted. The mAP, precision rate, recall rate and F1 score of proposed method are lower than those of Ref. [20, 33]. Ref [20, 33]. all use a single label deep hashing method based on an LSTM or CNN network model, which hash constructs the deep semantic information of the extracted MFCC and spectrogram features, and generates 512 bit and 384 bit binary hash codes with richer semantic information. However, the proposed method uses the triplet deep hash method based on the Tri-CNN. In the early stage of the model training, the loss is too large, resulting in the loss of part of the information, and the binary hash code of 128bit is used. In Ref. [20], the recall rate, precision rate and F1 score are the lowest when the spectrogram features are used as model inputs. This is because the LSTM network model structure is complex, and when the spectrogram features pass through LSTM, the loss is too large, resulting in serious loss of semantic information. The retrieval efficiency of proposed method is lower than that of Ref. [20, 33], because the retrieval process of proposed method is carried out in the FHE domain. Although the retrieval performance of the proposed method is not as good as that of the Ref. [20, 32, 33, 34], the experimental verification of proposed method from the perspective of spectrogram images is completed by using triplet deep hashing method, and its advantages in image field are fully exerted in the field of speech retrieval, and a more efficient and compact binary hash code is generated. At the same time, the secure speech retrieval under HE scheme is realized. Compared with the plaintext domain, the mAP, recall rate, precision rate and F1 score of proposed method lose about 2%, and the time delay is about 0.6 s. It was because CKKS FHE scheme encrypted the binary hash codes and constructed secure index. In order to ensure the security of the index, the polynomial module of 8,192 and the ciphertext module of 218 were used, so that each secure index involved ciphertext operations with four multiplication levels in the retrieval process. And because of the rescaling of the encryption scheme, the computation amount is relatively large. When the number of querying speech clips is 10,000, it involves 40,000 large integer ciphertext operations with multiplication levels, which affects the retrieval efficiency. Although the retrieval efficiency of proposed method in the encrypted domain is not high, the encryption scheme in this paper has solved the increasingly prominent privacy security problem, and realized high mAP, recall rate, precision rate and F1 score, thus meeting the needs of the speech security retrieval system.

4.5 Security analysis

4.5.1 *Correctness* analysis

For the public–private key pair (pk,sk) generated by KeyGen(λ), any plaintext m in plaintext domian M and encrypted C, if for $Enc(pk, m) \rightarrow c$, there is $Dec(sk, c) \approx m\prime$. The homomorphic encryption and decryption is considered to be correct.

Proof: Decryption correctness of ciphertext encrypted once.

If Eq. (23) holds, the decryption is correct.

$$\begin{aligned} m &= \langle c, sk \rangle mod Q \\ &= \langle (b, r + m + e_0, a \cdot r + e_1), (2, s) \rangle mod Q \\ &= ((-a \cdot s + e) \cdot r + m + e_0) + (a \cdot r \cdot s + e_1 \cdot s) mod Q \\ &= (e \cdot r + m + e_0 + e_1 \cdot s) mod Q \\ &\approx m\prime \end{aligned} \tag{23}$$

Because the rescaling of CKKS FHE algorithm can make the error $e \cdot r + e_0 + e_1 \cdot s$ smaller than Q, the error can be completely ignored. Therefore, the decryption of encrypted speech data is completely correct.

4.5.2 Analysis of speech retrieval security model

According to the security speech retrieval system model, because CS is "honest and curious", the user's private key is only stored locally, and CS is only responsible for storing data and security similarity measurement calculation, but cannot obtain the public and private center information of the user. The security definition of the semi trust model is as follows.

Definition (security of semi-trust model): Let function $p(x, y)$, where $p_1(x, y)$ and $p_2(x, y)$ are the first and second elements of $p(x, y)$, respectively. Suppose ϑ is a two-party protocol used to calculate $p(x, y)$. $CHALLENGE_1(x, y)$ is a role that implements the ϑ protocol, where $CHALLENGE_1(x, y) = (x, r, z_1, z_2, \ldots, z_t)$, where x represents input, r represents randomness, and z_i represents the i-th data received. And $CHALLENGE_1(x, y) = (y, r, z_1, z_2, \ldots, z_t)$ is also available. If there are probabilistic polynomial time algorithms V_1 and V_2, so that.

$$V_1 = (x, p_1(x, y)) \mapsto CHALLENGE_1(x, y)$$
$$V_2 = (y, p_2(x, y)) \mapsto CHALLENGE_2(x, y)$$

where the computational indistinguishability (IND-CPA) can be expressed as $\mapsto$, and when the ϑ protocol confronts a semi-trusted opponent, the computational $p(x, y)$ is secure.

Theorem 1: *Under the semi-honest model, the speech retrieval security model proposed in this paper is secure. CS can't get any useful information from stored data sets, deep binary hash codes and search trapdoors.*

Proof: In this protocol, the speech data, deep binary hash codes and search trapdoor that CS can obtain are transmitted in the form of ciphertext. The view of CS is $CHALLENGE = (t_1, t_2, \cdots, t_n)$, where t_i is all ciphertext and n is the number of ciphertext data that CS can access. The views of DO and DU can be simulated through simulator V, and $V = (r_1, r_2, \cdots, r_n)$ can be set, where r_i all random numbers is. Because CKKS FHE scheme is a semantic security encryption scheme, V is no different from $CHALLENGE$ in computing. Thus, under the semi honest model, the speech retrieval security model proposed in this paper can be proved to be security, and CS can't obtain any useful information from the stored speech data sets, deep binary hash codes, and search trapdoors.

Theorem 2: *Assume that attackers such as CS cannot perform key recovery attacks on the stored or stolen encrypted speech data and the encrypted speech data in the calculation results, so they cannot recover the user's original speech data and key.*

Proof: According to the protocol algorithm, the speech data, deep binary hash code and search trapdoor of DO and DU are transmitted in the form of CKKS ciphertext. The security of the retrieval method in this paper is protected by CKKS FHE scheme, which is granted with security and guaranteed by its algorithm. Thus, CS cannot recover the original

speech data and key through the stored encrypted speech data and intermediate calculation results. In the process of returning the results, the similarity calculation results are also transmitted to the DU in the form of ciphertext for decryption, so the attacker cannot recover the semantic information of the original speech data from the encrypted speech data intercepted in the transmission process and the stolen CS speech data.

4.5.3 Correlation analysis

Correlation coefficient is a statistical index reflecting the linear relationship between variables, which is often used to evaluate the encryption performance of encryption algorithms. The value range of the correlation coefficient r_{t} can be expressed as $r_t \in [-1, +1]$. If the value of the correlation coefficient of the two speech data is approximate to the boundary value, it means that the correlation between the two original speech data is very strong; If the correlation coefficient of the two speech data is approximately 0, it means that the correlation between the two original speech data is extremely weak. Equations (24)-(27) are expressions of correlation coefficients.

$$r_{\mathrm{t}} = \frac{C(j,o)}{\sqrt{\gamma(j)}\sqrt{\gamma(o)}} \tag{24}$$

$$C(j,o) = \frac{\sum(j-\bar{j})(j-\overline{o})}{N-1} \tag{25}$$

$$\gamma(m) = \frac{1}{N}\sum_{i=1}^{N}(j_i-\bar{j})^2 \tag{26}$$

$$\bar{j} = \frac{1}{N}\sum_{i=1}^{N} j_i \tag{27}$$

where $\bar{j}$ and $\overline{o}$ are respectively represented as the mean values of the original speech signal j and the encrypted speech signal o; $C(j, o)$ represents the covariance between the original speech signal j and the encrypted speech signal o; And $\gamma(j)$ and $\gamma(o)$ represent the variance between the original speech signal j and the encrypted speech signal o, respectively.

Since THCHS-30 has the same encryption effect as the speech data in TIMIT, only 4 s speech in TIMIT speech library is taken as an example. The feature comparison results of waveforms and spectrograms of original speech and encrypted speech are shown in Fig. 10.

As can be seen from Fig. 10 (a and b) , the waveform diagram of encrypted speech is evenly distributed, and no speech waveform features can be seen at all. It can be seen from Fig. 10 (c and d) that the spectrogram image features of encrypted speech are very similar to white noise, and there is no energy feature. Thus, the encryption scheme in this paper has high security, is very suitable for speech data encryption, and the effect is good. In order to evaluate the anti-statistical analysis attack performance of the encryption scheme in this paper, the correlation coefficient of encrypted signals is calculated for analysis, and Pearson correlation coefficient of speech signals is calculated according to Eq. (24) to measure the correlation between speech signals. The calculation results are shown in Table 8.

It can be seen from Table 8 that the correlation coefficient between the original speech and the encrypted speech is approximately 0 under the THCHS-30 and TIMIT data sets, which indicates that the original speech and the encrypted speech have no correlation

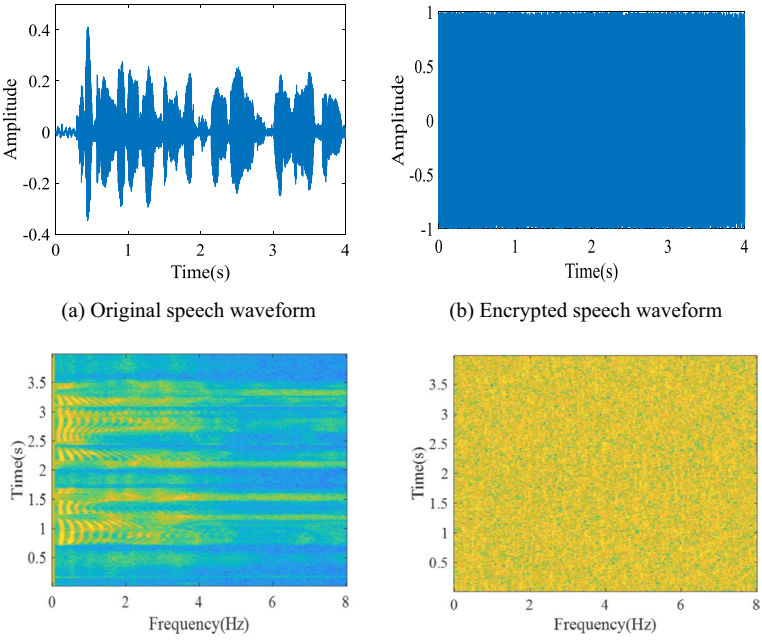
and the encryption performance is good. The correlation coefficient between the original speech and the decrypted speech is approximately 1, which means that the decryption algorithm is good and can basically recover the speech signal without loss.

4.5.4 Encryption and decryption efficiency analysis

The total number of pre-processed speech data is 64,000 items. If the direct encryption needs to perform 64,000 encryption algorithms, 64,000 encrypted speeches will be obtained. The encryption scheme in this paper is to batch the pre-processed speech data, package it into 8 plaintext matrices, and batch the 8 plaintext matrices when encrypting, so as to obtain 8 encrypted speeches. Table 9 shows the average time consumption obtained by encrypting and decrypting the speech data in THCHS-30(4 s) and TIMIT(4 s) speech data sets for 8 times in this speech encryption scheme.

As can be seen from Table 9, the decryption speed of the encryption scheme in this paper is about 3 times faster than that of encryption in the data sets of THCHS-30(4 s) and TIMIT(4 s). This is because the algorithm needs to be rescaled and the numerical value is enlarged before encoding the data, which makes the calculation amount very large. However, decryption is the reverse process of encoding, which reduces the subsequent operation scale and saves a lot of time.

Figure 11 shows the time taken by the encryption scheme in this paper to construct secure indexes of different lengths under the THCHS-30 and TIMIT data sets.



(a) Original speech waveform (b) Encrypted speech waveform

(c) Original speech spectrogram (d) Encrypted speech spectrogram

Fig. 10 Comparison of waveform and spectrogram image features between original speech and encrypted speech

Table 8 Speech Correlation Analysis

Data set	Speech file name	Original speech & Encrypted speech	Original speech & Decrypt speech
THCHS-30	32kbps_A9_5.wav	0.0032	0.97969
	50%_A19_8.wav	0.0031	0.98876
	48kbps_A4_0.wav	0.0028	0.98285
	average value	0.0030	0.98377
TIMIT	32kbps_Sa1_0.wav	0.0032	0.97989
	50%_Sa1_5.wav	0.0031	0.99985
	48kbps_Sa1_9.wav	0.0027	0.98456
	average value	0.0030	0.98810

As can be seen from Fig. 11, the time consumption of constructing a secure index with a length of 64, 128, 256 shows a linear increase, and the longer the length, the more speech clips and the more time spent. When the number of speech clips is 1,000, it takes 1.3107 s, 2.1485 s, and 4.1622 s to construct secure index for 64 bit, 128 bit, and 256 bit, respectively.

Table 10 shows the time taken by the encryption scheme in this paper to encrypt search trapdoors with different lengths under THCHS-30 and TIMIT data sets.

As can be seen from Table 10, when the encryption scheme in this paper encrypts 64 bit, 128 bit and 256 bit search trapdoor, it takes less than 1 s, and it can be completed almost instantly, which shows that the time cost of encrypting search trapdoor is basically negligible.

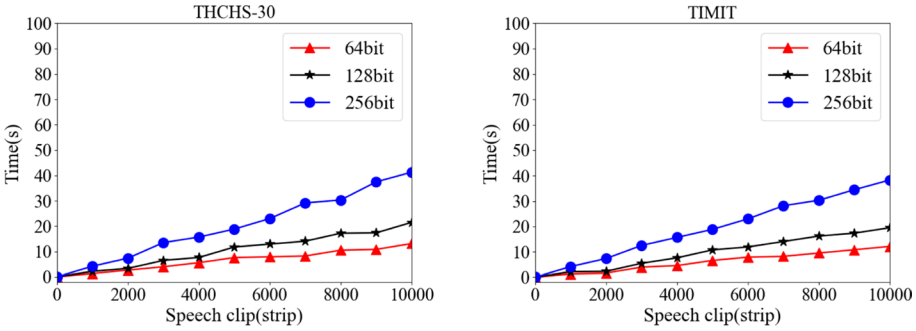
4.5.5 Comprehensive performance analysis of different encryption schemes

In order to objectively evaluate the performance of the encryption scheme in this paper, its performance is compared with the improved speech HE scheme [35–37]. As the encryption and decryption effects of the encryption scheme in this paper on THCHS-30 and TIMIT datasets are similar, the 4 s speech data in TIMIT is taken as an example. Table 11 shows the performance comparison between this encryption scheme and other encryption schemes.

It can be seen from Table 11 that the overall performance of the encryption scheme in this paper is better than the encryption scheme in [35, 36]. The larger the key size and key space, the higher the security performance of the algorithm. The security performance of the encryption schemes in this paper is smaller than that of Ref. [35–37], because the encryption schemes in this paper deal with complex numbers, and its security depends on polynomial degree N and secret text module q. To enhance the security of the algorithm, it is necessary to increase the value of q, but the only way to increase the size of q is

Table 9 Encryption and decryption time consumption of CKKS scheme

Algorithm	Calculation time (s)	
	THCHS-30	TIMIT
Encryption	16.7859421	16.7838947
Decryption	5.5210688	5.5112472



(a) THCHS-30 (b) TIMIT

Fig. 11 The time consuming comparison of building the secure index

Table 10 Time spent encrypting search trapdoors with different lengths under different data sets

Data set	THCHS-30			TIMIT		
Hash code length (bit)	64	128	256	64	128	256
Encryption time (s)	0.00131	0.00215	0.00416	0.00131	0.00215	0.00416

Table 11 Performance comparison between this encryption scheme and other encryption schemes

Evaluation Indicators	Proposed	Ref. [35]	Ref. [36]	Ref. [37]
Key size	218	512	512	592
Key space	2^{218}	2^{512}	2^{512}	2^{592}
Cov (Original & Encryption)	0.0030	0.0004	0.0746	0.0003
Cov (Original & decryption)	0.9838	0.8006	1.0000	1.0000
Encryption time (s)	16.7859	1.1240	25.3075	7.1456
Decryption time (s)	5.5211	0.6240	24.2481	3.5345
Ciphertext extension	2.255×10^{3}	5.7×10^{4}	6.667×10^{6}	2.521×10^{4}

to increase the size of N, which will seriously increase the computational complexity and greatly reduce the efficiency of encryption and decryption. Although the security performance of the encryption scheme in this paper is not the highest, the polynomial degree N and the secret text module q used have met the needs of users' privacy data security. The closer the correlation between the original speech data and encrypted speech data is to 0, the better the security performance of the encryption scheme. The Cov (original & encryption) of the encryption scheme in this paper is smaller than that in Ref. [36], and only about 0.0026 less than that in Ref. [35, 37], indicating that the encryption scheme in this paper is highly secure and the attacker cannot obtain data privacy information. The closer the correlation between the original speech data and the decrypted speech data is to 1, the higher the decryption quality of the decryption algorithm. The Cov (original & decryption) of the encryption scheme in this paper is larger than that in Ref. [35], and is not much different from that in Ref. [36, 37], only about 0.01623. This shows that the decryption quality of the

decryption algorithm in this paper can meet the user's requirements, and the undistorted decrypted speech data can be basically obtained. This difference is because the encryption scheme in this paper is an approximate calculation of data, and there are some errors in the calculation process, but this error can be ignored. The encryption efficiency and decryption efficiency of the encryption scheme in this paper are 1.5 times and 4 times faster than that of the probabilistic plus homomorphic cryptosystem proposed in Ref. [36]. This is because the encryption scheme in this paper uses CRT and SIMD batch processing technology to package data onto eight matrices and implement cyclic encryption. The encryption times are reduced from 64,000 to 8, significantly improving the encryption and decryption time. Compared with Ref. [35, 37], the encryption efficiency and decryption efficiency of the encryption scheme in this paper are slightly lower. This is because the encryption scheme in this paper is FHE scheme and deals with complex numbers, and the rescaling operation enlarges the numerical value in the encryption process, making its operation scale very large. However, the method in Ref. [35, 37] only deals with integers, so the calculation amount is small, and Ref. [37] uses the multithreading technology. Although the encryption time and decryption time of the encryption scheme in this paper are not optimal, the encryption scheme in this paper can handle complex numbers and has a wider application range. The ciphertext expansion of the encryption scheme in this paper is less than that in the Ref. [35–37], which is because the encryption scheme in this paper uses a speech matrix encryption method, which reduces the interactive information of encrypted speech data in the encryption process, limits the expansion of data, and uses modular switching technology and relinearization technology to reduce the ciphertext dimension, reduce the ciphertext expansion, and reduce memory consumption. Thus, the encryption scheme in this paper is very suitable for cloud secure speech recognition, secure speech retrieval, etc., which can better realize the security sharing and privacy protection of data, and has high security.

5 Conclusions

In order to solve the problems of secure storage, secure sharing and fast retrieval of massive speech data in cloud environment, a secure speech retrieval method based on deep hashing and CKKS FHE is proposed. The proposed method solves the problem that the deep binary hash codes is easy to be attacked during the retrieval of encrypted speech data in cloud storage, resulting in data leakage, while keeping the retrieval performance basically unchanged. The proposed method adopts triple deep hashing method, fully extracts the deep semantic information of spectrogram image features by using Tri-CNN, and generates the deep binary hash codes with maximum discrimination, which reduces the matching dimension of deep binary hash codes and improves the efficiency of secure speech retrieval to a certain extent. In order to ensure the secure sharing of data, CKKS FHE scheme is used to protect the privacy of the original speech data and the deep binary hash codes, and then the cosine similarity is used for matching retrieval, thus realizing the secure speech retrieval with privacy protection. The experimental results of TIMIT and THCHS-30 data sets show that the proposed method can satisfy the inseparability of secure index and trapdoor information, realize the secure sharing of data, and achieve high retrieval performance.

As a future work, we will further study the secure speech retrieval method that supports dynamic update of secure index table in cloud storage.

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Data availability Previously reported speech data (THCHS-30 and TIMIT data sets) were used to support this study and are available at 10.48550/arXiv.1512.01882 and 10.1016/0167–6393(90)90,010–7. This is cited at relevant places within the text as reference [25, 26].

Declarations

Conflict of interests The authors declare that they have no conflict of interests.

References

1. Li Y, Ma J, Miao Y et al (2022) Similarity search for encrypted images in secure cloud computing[J]. IEEE Transactions on Cloud Computing 10(2):1142–1155. https://doi.org/10.1109/TCC.2020.2989923
2. Singh N, Kumar J, Singh A K, et al. Privacy-preserving multi-keyword hybrid search over encrypted data in cloud[J]. Journal of Ambient Intelligence and Humanized Computing, 2022: 1–14. https://doi.org/10.1007/s12652-022-03889-8
3. Rahulamathavan Y. Privacy-preserving Similarity Calculation of Speaker Features Using Fully Homomorphic Encryption[J]. arXiv preprint arXiv:2202.07994, 2022. https://doi.org/10.48550/arXiv.2202.07994
4. Shen M, Cheng G, Zhu L et al (2020) Content-based multi-source encrypted image retrieval in clouds with privacy preservation[J]. Futur Gener Comput Syst 109:621–632. https://doi.org/10.1016/j.future.2018.04.089
5. Duan Y, Li Y, Lu L et al (2022) A faster outsourced medical image retrieval scheme with privacy preservation[J]. J Syst Archit 122:102356. https://doi.org/10.1016/j.sysarc.2021.102356
6. Wang Q, Feng C, Xu Y et al (2020) A novel privacy-preserving speech recognition framework using bidirectional LSTM[J]. Journal of Cloud Computing 9(1):1–13. https://doi.org/10.1186/s13677-020-00186-7
7. Shi C, Wang H, Hu Y et al (2021) A novel NMF-based authentication scheme for encrypted speech in cloud computing[J]. Multimedia Tools and Applications 80(17):25773–25798. https://doi.org/10.1007/s11042-021-10896-y
8. Wang Y, Huang Y, Zhang R et al (2021) Multi-format speech BioHashing based on energy to zero ratio and improved LP-MMSE parameter fusion[J]. Multimedia Tools and Applications 80(7):10013–10036. https://doi.org/10.1007/s11042-020-09701-z
9. Zhang S X, Gong Y, Yu D. Encrypted Speech Recognition Using Deep Polynomial Networks[C]. ICASSP 2019–2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP).IEEE, Brighton, 2019:5691–5695. https://doi.org/10.1109/ICASSP.2019.8683721
10. Yu X, Xu C, Dou B et al (2021) Multi-user search on the encrypted multimedia database: lattice-based searchable encryption scheme with time-controlled proxy re-encryption[J]. Multimedia Tools and Applications 80(2):3193–3211. https://doi.org/10.1007/s11042-020-09753-1
11. Cao R, Zhang Q, Zhu J et al (2020) Enhancing remote sensing image retrieval using a triplet deep metric learning network[J]. Int J Remote Sens 41(2):740–751. https://doi.org/10.1080/2150704X.2019.1647368
12. Li M, An Z, Wei Q et al (2019) Triplet Deep Hashing with Joint Supervised Loss Based on Deep Neural Networks[J]. Comput Intell Neurosci 2019:1–17. https://doi.org/10.1155/2019/8490364
13. Jia Y, Chen X, Yu J et al (2021) Speaker recognition based on characteristic spectrograms and an improved self-organizing feature map neural network[J]. Complex & Intelligent Systems 7(4):1749–1757. https://doi.org/10.1007/s40747-020-00172-1
14. Purwins H, Li B, Virtanen T et al (2019) Deep Learning for Audio Signal Processing[J]. IEEE J Selected Top Signal Process 13(2):206–219. https://doi.org/10.1109/JSTSP.2019.2908700
15. Cheon J H, Kim A, Kim M, et al. Homomorphic encryption for arithmetic of approximate numbers[C]//International conference on the theory and application of cryptology and information security. Springer, Cham, 2017: 409–437. https://doi.org/10.1007/978-3-319-70694-8_15
16. Chen C, Jiang D, Peng J et al (2021) Scalable Identity-Oriented Speech Retrieval[J]. IEEE Trans Knowl Data Eng 14(8):1–6. https://doi.org/10.1109/TKDE.2021.3127520
17. Zhang Q, Li Y, Hu Y (2021) A retrieval algorithm for encrypted speech based on convolutional neural network and deep hashing[J]. Multimedia Tools and Applications 80(1):1201–1221. https://doi.org/10.1007/s11042-020-09748-y

18. Zhang H (2021) Voice keyword retrieval method using attention mechanism and multimodal information fusion[J]. Sci Program 2021(8):1–11. https://doi.org/10.1155/2021/6662841
19. Yuan Y, Xie L, Leung CC et al (2020) Fast query-by-example speech search using attention-based deep binary embeddings[J]. IEEE/ACM Transactions on Audio, Speech, and Language Processing 28:1988–2000. https://doi.org/10.1109/TASLP.2020.2998277
20. Zhang Q, Fu M, Huang Y et al (2022) Encrypted Speech Retrieval Scheme Based on Multiuser Searchable Encryption in Cloud Storage[J]. Security and Communication Networks 2022:9045259. https://doi.org/10.1155/2022/9045259
21. Li W, Chen Y, Hu H et al (2020) Using granule to search privacy preserving voice in home IoT systems[J]. IEEE Access 8:31957–31969. https://doi.org/10.1109/ACCESS.2020.2972975
22. Li W, Xiao Y, Tang C et al (2020) Multi-user searchable encryption voice in home IoT system[J]. Internet of Things 11:100180. https://doi.org/10.1016/j.iot.2020.100180
23. Chen J, Chen Z, Zheng P, et al. Encrypted domain mel-frequency cepstral coefficient and fragile audiowatermarking[C]. 2018 17th IEEE International Conference On Trust, Security And Privacy In Computing And Communications/12th IEEE International Conference On Big Data Science And Engineering (TrustCom/BigDataSE). IEEE, New York, 2018: 68-73. https://doi.org/10.1109/TrustCom/BigDataSE.2018.00021
24. Thaine P, Penn G (2019) Extracting Mel-Frequency and Bark-Frequency Cepstral Coefficients from Encrypted Signals[C]. INTERSPEECH, Graz 3715–3719. https://doi.org/10.21437/Interspeech.2019-1136
25. Tang Y, Zhu B, Ma X, et al (2019) Decoding homomorphically encrypted FLAC audio without decryption[C]//ICASSP 2019–2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE 675–679. https://doi.org/10.1109/ICASSP.2019.8682780
26. Meftah S, Tan BHM, Mun CF et al (2021) Doren: toward efficient deep convolutional neural networks with fully homomorphic encryption[J]. IEEE Trans Inf Forensics Secur 16:3740–3752. https://doi.org/10.1109/TIFS.2021.3090959
27. Natarajan D, Dalskov A, Kales D, et al (2021) PRIORIS: Enabling Secure Detection of Suicidal Ideation from Speech Using Homomorphic Encryption[M]//Protecting Privacy through Homomorphic Encryption. Springer, Cham 133–146. https://doi.org/10.1007/978-3-030-77287-1_10
28. Liu J, Wang C, Tu Z et al (2021) Secure KNN classification scheme based on homomorphic encryption for cyberspace[J]. Secur Commun Netw 2021:8759922. https://doi.org/10.1155/2021/8759922
29. Wang D, Zhang X (2015) Thchs-30: A free chinese speech corpus[J]. arXiv preprint arXiv: 1512.01882. https://doi.org/10.48550/arXiv.1512.01882
30. Zue V, Seneff S, Glass J (1990) Speech database development at MIT: TIMIT and beyond[J]. Speech Commun 9(4):351–356. https://doi.org/10.1016/0167-6393(90)90010-7
31. Ullah B, Kamran M, Rui Y (2022) Predictive modeling of short-term rockburst for the stability of subsurface structures using machine learning approaches: T-SNE, K-Means clustering and XGBoost[J]. Mathematics 10(3):449. https://doi.org/10.3390/math10030449
32. An L, Huang Y, Zhang Q (2022) Verifiable speech retrieval algorithm based on KNN secure hashing[J]. Multimedia Tools and Applications 1–22. https://doi.org/10.1007/s11042-022-13387-w
33. Zhang Q, Zhao X, Zhang Q et al (2022) Content-based encrypted speech retrieval scheme with deep hashing[J]. Multimed Tools Appl 81(7):10221–10242. https://doi.org/10.1007/s11042-022-12123-8
34. Huang Y, Wang Y, Li H et al (2022) Encrypted speech retrieval based on long sequence Biohashing[J]. Multimed Tools Appl 81(9):13065–13085. https://doi.org/10.1007/s11042-022-12371-8
35. Khoirom MS, Laiphrakpam DS, Tuithung T (2021) Audio encryption using ameliorated ElGamal public key encryption over finite field[J]. Wireless Pers Commun 117(2):809–823. https://doi.org/10.1007/s11277-020-07897-9
36. Shi C, Wang H, Hu Y et al (2019) A Speech Homomorphic Encryption Scheme with Less Data Expansion in Cloud Computing[J]. KSII Trans Internet Inf Syst (TIIS) 13(5):2588–2609. https://doi.org/10.3837/tiis.2019.05.020
37. Zhang QY, Jia YG (2022) A Speech Fully Homomorphic Encryption Scheme for DGHV Based on Multithreading in Cloud Storage [J]. Int J Netw Secur 24(6):1042–1055. https://doi.org/10.6633/IJNS.202211124(6).09

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